Target Detection for Multistatic Radar in the Presence of Deception Jamming

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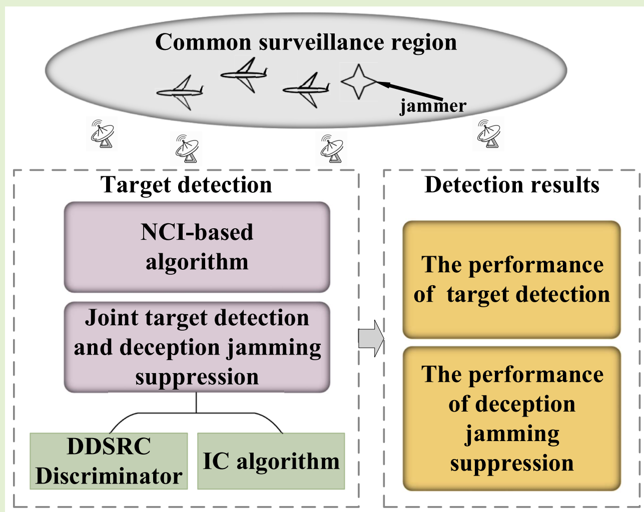
***Abstract*—This study investigates the target detection problem of multistatic radar in the presence of deception jamming. A detection algorithm based on non-coherent integration is presented to determine whether a target exists in each probed spatial resolution cell (SRC). This algorithm can suppress the jamming to some extent because the spatial divergence of the jamming signal inhibits the effective energy accumulation for detection. However, a jamming signal influences more SRCs than a target. Given that the power of deception jamming is usually larger than that of the target, the global statistics of the SRCs can still exceed the detection threshold with a certain probability, which leads to many false targets. Therefore, a discriminator is presented to distinguish the jamming signals from the target echoes and noises. Subsequently, a joint target detection and deception jamming suppression method that can simultaneously detect targets and suppress the jamming is derived to eliminate the interference in the relative SRCs. Simulation results show that the proposed algorithm can suppress the deception jamming while accurately detecting multiple targets.**



***Index Terms*—Multistatic radar, multi-target detection, deception jamming discrimination, spatial resolution cell, interference cancellation.**

I. Introduction

MULTISTATIC radar is defined as a comprehensive radar system that includes several spatially separated transmitting and receiving facilities where the observation data from different radar sites can be fused and jointly processed [1]. The multistatic radar with long baseline observes a common surveillance region from different angles to allow the system to utilize the spatial diversity of the target scatterers. Various new techniques are applied to this radar configuration to enhance the target detection [2], [3], target localization [4], [5], and electronic counter-countermeasure [6], [7].

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The target detection performance of multistatic radar is better than that of monostatic radar, particularly for weak targets, because of the spatial diversity gain [8]–[10]. The detection problem is to determine whether a target is present in the probed spatial resolution cell (SRC) [1]. Several target detection methods for multistatic radar system (MSRS) were proposed. These methods can be divided into two categories in accordance with the fusion level in the fusion center. The first is distributed target detection, which is a two-step method in which each radar station uses its own observation data to make decisions and then fuses the local decision or statistic in the signal fusion center for further decision-making [11]–[13]. This approach is usually accompanied by performance loss because the local thresholding filters the target information.

The second category is centralized target detection. The methods under this category make decisions regarding the presence of a target in the signal fusion center by comparing the global test statistic and a predetermined threshold. The global test statistic is calculated by fusing the received signals in the fusion center. These techniques are one-step target detection approaches. Given the absence of local thresholding in each radar station, these approaches suffer from minimal information loss. Many centralized target detection algorithms have been proposed in different studies. Several optimum target detectors for multistatic radar derived in the background of self-noises in [1], [2], [14]. Taking clutter into consideration, the target detection algorithms based on the generalized likelihood ratio test algorithm were proposed [15], [16]. These algorithms focus on the optimum detection in an isolated SRC but disregard the fact that a target can affect more than one SRC. As a result, the false alarm probability in the neighboring SRCs increases. To address this problem, a deep insight into the centralized target detection in multistatic radar was presented in [17], [18]. In [17] a general scheme of the SRC-based centralized target detection was proposed. The peak search method is used to find the SRC that contains the target. However, this approach does not consider the existence of multiple targets in the common surveillance region. Errors in data association may cause a "ghost target" problem [19]. To avoid this issue, a joint multi-target detection and localization method was proposed [18]. The abovementioned algorithms focus on the problem of optimum detection with noise or noise-like interference backgrounds and assume that the SRC in the common surveillance area can only contain noise and the target. However, radar systems are always exposed to complicated electromagnetic environments where various jamming instances, especially active deception jamming, degrade the radar performance. This condition complicates the detection problem. MSRSs can detect many false targets when the hostile deception jammer retransmits the intercepted radar signals after appropriate modulations [20]. This phenomenon occurs because the deception jamming signals can exceed the test threshold with a certain probability during centralized target detection. If the jamming is not suppressed, the radar system will be confused, and the target tracking system will be saturated.

To eliminate deception jamming, physical and false targets are discriminated after the target detection. Several works adopted this approach. Some methods discriminate the false target with the measurements (e.g., range, angle, and Doppler information) obtained by local radar sites because the measurements from different radars are generally not equal for radar target echoes, but equal for deception signals [21], [22]. Several discriminators are based on the differences in spatial scattering properties between the jamming and target signals [23]–[27]. These discrimination-based methods require the extraction of the features of each potential target from the observation data gathered by different radar sites. Moreover, these methods assume that there are no errors in the data association. Errors in data association will lead to a severe ghost target problem because more potential targets will be affected by deception jamming. The features of a ghost target may contain noise-only components, which inhibits correct discrimination.

In this study, we analyze the mechanism of centralized target detection in the presence of deception jamming. A detection algorithm based on non-coherent integration (NCI) is introduced, and the influences exerted by deception jamming on target detection are analyzed. Given that the power of the jamming signals is usually larger than that of the target, these signals are still detected by mistake despite their loss of effective energy accumulation during detection. Subsequently, a joint target detection and deception jamming suppression method is presented. The method includes two blocks, namely, the discrimination of deception-dominant SRC (DDSRC) and the interference cancellation (IC) algorithm; the former is the prerequisite of the latter. The proposed algorithm derives an extra term to modify the NCI-based algorithm. Given that the modified term may be updated once a target or a jamming signal is declared, the threshold should be adjusted accordingly. Numerical examples are provided to assess the detection and jamming suppression performances of the proposed algorithms.

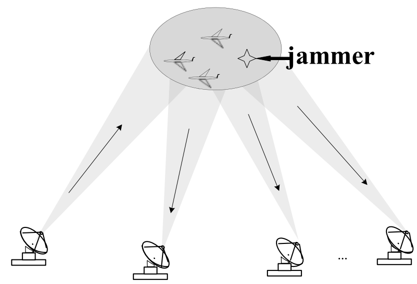

Fig. 1. Sketch of spatial configuration.

The remainder of the paper is organized as follows. Section II introduces the model and the notations. Section III discusses the target detection performance in the presence of deception jamming and analyzes the difference between the target echo and the jamming signal by using the NCI-based algorithm. Section IV presents the proposed joint target detection and deception jamming suppression method. Section V discusses the experimental results. Lastly, Section VI provides the conclusions.

II. Models and Notations

We consider an MSRS with a transmitter located at (x_0^t, y_0^t) and M receivers located at (x_m^r, y_m^r), where $m = 1, 2, 3, \ldots, M$. The antennas of the transmitter and receivers, which are assumed to already have time synchronization [28]–[30], are widely separated to provide space diversity gain. The transmitter and the receivers simultaneously observe a common surveillance region denoted by Ω. K $(K > 1)$ targets are located at (x_k^p, y_k^p), where $k = 1, 2, \ldots, K$. To protect the targets from the radar system, a repeater jammer is placed at (x_q^j, y_q^j). Fig. 1 illustrates the spatial configuration of the radar system, the targets, and the jammer. Considering the interference of the jammer, the digital signal received by the mth receiver is expressed as

$$r_m[n] = \sum_{k=1}^{K} e_{m,k}[n] + \sum_{q=1}^{Q} j_{m,q}[n] + \upsilon_m[n]\frac{1}{2}, \tag{1}$$

where $n = 0, 1, \ldots, N-1$. $N = \lfloor PRT/T_s + 0.5 \rfloor$ is the length of the discrete signal with the sampling interval of the receivers T_s and the floor operation $\lfloor \cdot \rfloor$. $e_{m,k}[n]$ is the ideal echo reflected by the kth target, $j_{m,q}[n]$ is the qth deception jamming signal, and $\upsilon_m[n]$ represents the additive complex white Gaussian random noise with mean 0 and variance. The noise levels of all receivers are assumed to be equal, that is,

$\sigma_1 = \sigma_2 = \ldots = \sigma_m = \sigma$. The transmitted signal is denoted as $s(t)$. By defining $s_{m,k}[n; \tau] \cdot s_{m,k}(nT_s - \tau)$, the ideal target echo can be written as

$$e_{m,k}[n] = \alpha_{m,k} s\left[n; \tau_{m,k}^{p}\right], \tag{2}$$

where $\alpha_{m,k}$ is the complex amplitude of the target echo, which is related to the transmitting power, propagation loss, radar cross section, and radial velocity of the target, and $\tau_{m,k}^{p} = h(x_k^p, y_k^p)$ denotes the radar signal delay from the transmitter to the mth receiving antenna reflected by the kth target. The function $h(\cdot)$ can be expressed as

$$h(x, y) = \frac{1}{c}\left[\sqrt{(x - x^t)^2 + (y - y^t)^2} + \sqrt{(x - x_m^r)^2 + (y - y_m^r)^2}\right], \tag{3}$$

where c is the speed of light.

A point-like target is assumed to facilitate the acquisition of the independent observations of each target [31].

$$\sum\nolimits_N s[n; \tau_i] s^*[n; \tau_j] = \begin{cases} 1 & \left|\tau_i - \tau_j\right| < 1/2B \\ 0 & \left|\tau_i - \tau_j\right| > 1/2B \end{cases} \tag{4}$$

where B is the signal bandwidth.

The deception jamming signal can be defined as

$$j_{m,q}(t) = \beta_{m,q}\left[n; \tau_{m,q}^{j} + \tilde{\tau}\right], \tag{5}$$

where $\beta_{m,q}$ denotes the complex amplitude of the signal retransmitted by the repeater jammer, $\tau_{m,q}^{j} = h(x_q^j, y_q^j)$ represents the delay from the transmitter to the mth receiving antenna reflected by the jammer, and $\tilde{\tau}_q$ is the time delay modulation of the jammer.

Equation (1) can be written in a vector form as

$$\mathbf{r}_m = \sum_{k=1}^{K} \mathbf{e}_{m,k} + \sum_{q=1}^{Q} \mathbf{j}_{m,q} + \boldsymbol{\upsilon}_m, \tag{6}$$

where $\mathbf{r}_m = [r_m\ [0], r_m\ [1], \ldots, r_m[N-1]]^T$ denotes the vector form of the receiving signal with matrix transpose $[\cdot]^T$ and $\mathbf{e}_{m,k}, \mathbf{j}_{m,q}, \boldsymbol{\upsilon}_m$ is the vector form of the ideal target echo, jamming signal, noise received by the mth receiver.

The local signal $\mathbf{r}_m$ is gathered by the mth receiver based on its own coordinate reference system. To integrate multiple radar receiving signals into a single air picture, each signal of the local radar must be expressed in a common coordinate system [32]–[34]. Therefore, the local signals should be transformed into observation vectors of SRCs.

The term $c(\boldsymbol{\theta}_g) \in \boldsymbol{\Omega}$ denotes one of the SRCs of the common surveillance region with geometric center $\boldsymbol{\theta}_g = (x_g^c, y_g^c)$, where $g = 1, 2, \ldots, G\ (G > 0)$. G denotes the total number of SRCs in the common surveillance region. The common surveillance region is assumed to be covered by these SRCs [35]. The space mapping from the signal in the local coordinate reference system to the observation data in the common coordinate reference system is defined as

$$z_m(\boldsymbol{\theta}_g) = \tilde{\mathbf{s}}_m(\boldsymbol{\theta}_g)^H \mathbf{r}_m, \tag{7}$$

where $z_m(\boldsymbol{\theta}_g)$ is the observation data of the SRC gathered by the mth receiver, $(\cdot)^H$ represents the complex conjugate transpose operation, and $\tilde{\mathbf{s}}_m(\boldsymbol{\theta}_g)$ is the function of $\boldsymbol{\theta}_g$ that represents the reference vector of the SRC under detection. This function can be defined as

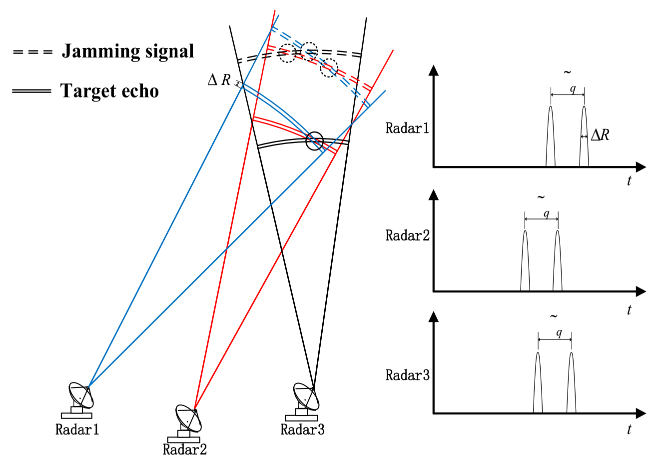

Fig. 2. Sketch of a scenario with a target echo and a deception jamming signal.

$$\tilde{\mathbf{s}}_m(\boldsymbol{\theta}_g) = \begin{bmatrix} s\left[0; \tau_m(\boldsymbol{\theta}_g)\right] \\ s\left[1; \tau_m(\boldsymbol{\theta}_g)\right] \\ \vdots \\ s\left[N-1; \tau_m(\boldsymbol{\theta}_g)\right] \end{bmatrix}, \tag{8}$$

where $\tau_m(\boldsymbol{\theta}_g) = h\left(x_g^c, y_g^c\right)$ is the time delay from the gth SRC at the mth receiving station. The observation vector gathered in the SRC is expressed as

$$\mathbf{z}\left(\boldsymbol{\theta}_g\right) = \left[z_1(\boldsymbol{\theta}_g), z_2(\boldsymbol{\theta}_g), \ldots, z_M(\boldsymbol{\theta}_g)\right]^T. \tag{9}$$

For a given SRC $c\left(\boldsymbol{\theta}_g\right)$, the relative SRCs from the -mth branch are defined as

$$\boldsymbol{\Psi}_m(\boldsymbol{\theta}_g) = \left\{c\left(\boldsymbol{\theta}^{re}\right) \middle| \left|\tau_m\left(\boldsymbol{\theta}_g\right) - \tau_m\left(\boldsymbol{\theta}^{re}\right)\right| < 1/2B, \boldsymbol{\theta}^{re} \neq \boldsymbol{\theta}_g\right\}. \tag{10}$$

III. Target Detection in the Presence of Deception Jamming

The detection problem in the environment with deception jamming might be complicated. As shown in Fig. 2, the deception jamming signals and target echoes can be gathered in the SRCs. Nevertheless, the characteristics of the SRCs affected by both factors are different.

A. Observation Vector Analysis for the Target

Assuming that the perfect registration is implemented, the local resolution cell of a target measured by a different radar station can intersect in the target-presence SRC after being transformed into a common coordinate system [36]. This intersection improves the detection performances.

The observation data of the kth target can be gathered by the mth receiving station in the SRC (i.e., $z_m(\boldsymbol{\theta}_g) = \alpha_{m,k} + \omega_m$, $\omega_m = \tilde{\mathbf{s}}_m(\boldsymbol{\theta}_g)^H \mathbf{n}_m$) if the time difference of arrival between the target and the SRC is smaller than the pulse width [18].

$$\boldsymbol{\Theta}_{m,k} = \left\{c(\boldsymbol{\theta}_g) \middle| \left|\tau_m(\boldsymbol{\theta}_g) - \tau_{m,k}^{p}\right| < 1/2B\right\} \tag{11}$$

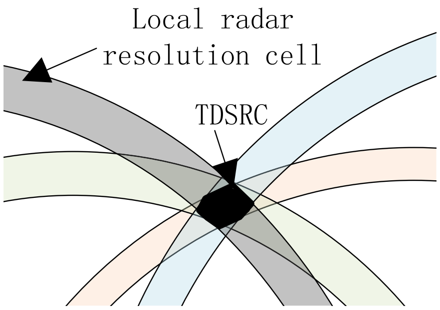

Fig. 3. Dominant and relative SRCs of a target.

The SRC containing the kth target is defined as thetarget-dominant SRC (TDSRC). The spatial power superposition of the target is the largest in TDSRC. Such SRCs can be expressed as

$$\mathbf{T}_k^{in} = \bigcap_{m=1}^{M} \mathbf{\Theta}_{m,k}. \tag{12}$$

In Fig. 3, the strips denote the resolution cells of a local radar whose intersection is the TDSRC, and the union sets of the local radar resolution cells, except for TDSRC, are the relative SRCs of a target.

For the TDSRC of the kth target (i.e., $c(\boldsymbol{\theta}_k^T)$), the observation vector can be expressed as

$$\mathbf{z}(\boldsymbol{\theta}_k^T) = \boldsymbol{\alpha}_k + \boldsymbol{\omega} = \left[\alpha_{1,k} + \omega_1, \alpha_{2,k} + \omega_2, \ldots, \alpha_{M,k} + \omega_M\right]^T. \tag{13}$$

where $\boldsymbol{\alpha}_k = \left[\alpha_{1,k}, \alpha_{2,k}, \ldots, \alpha_{M,k}\right]^T$ and $\boldsymbol{\omega} = [\omega_1, \omega_2, \ldots, \omega_M]^T$.

B. Effect of Deception Jamming

As previously mentioned, although deception jamming signals and target echoes can be gathered in the SRCs, their characteristics are different from each other.

If the observation data of the qth jamming signal can be gathered by the mth receiving station in the SRC (i.e. $z_m(\boldsymbol{\theta}_g) = \beta_{m,q} + \omega_m$),

$$\mathbf{\Phi}_{m,q} = \left\{c(\boldsymbol{\theta}_g) \,\middle|\, \left|\tau_{m,g}^c - \tau_{m,q}^j - \tilde{\tau}\right| < 1/2B\right\}. \tag{14}$$

The DDSRC is defined as the SRC where the most jamming signals can be collected by the multistatic radar. Deception jammers can hardly generate deception jamming signals with matched range relationship to each station [22], [26]. The measured local resolution cell of a jamming signal cannot intersect with an SRC even without the registration error

As shown in Fig. 4, every two local radar resolution cells of the deception jamming signal can intersect each other, but none of them has common overlaps. The overlapping intersections are the DDSRCs. The local radar resolution cells, except for the DDSRCs of the jamming signal, are the relative SRCs.

$$\forall a \neq b \neq c \quad a, b, c \in \{1, 2, \ldots, M\}$$
$$\mathbf{\Phi}_{a,q} \cap \mathbf{\Phi}_{b,q} \cap \mathbf{\Phi}_{c,q} = \emptyset \tag{15}$$

The DDSRCs where the qth jamming signal is gathered by the ith and jth receivers can be denoted as $\mathbf{J}_{i,j,q}^{in} = \mathbf{\Phi}_{i,q} \cap \mathbf{\Phi}_{j,q}$,

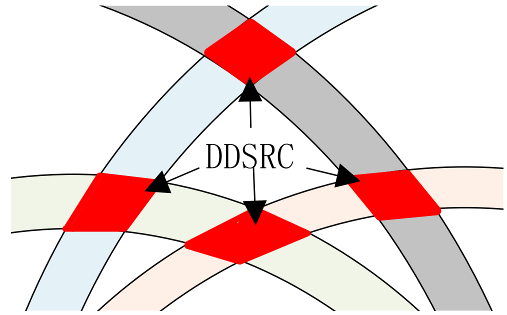

Fig. 4. DDSRCs of a jamming signal.

where $i, j \in\in \{1, 2, \ldots, \mathrm{M}\}$. For the DDSRC denoted as $c(\boldsymbol{\theta}_{i,j,q}^{jam})$ the observation vector can be written as

$$\begin{aligned}\mathbf{z}(\boldsymbol{\theta}_{i,j,q}^{jam}) &= \boldsymbol{\beta}_{i,j,q} + \boldsymbol{\omega} \\ &= \left[\omega_1, \ldots, \beta_{i,q} + \omega_i, \ldots, \beta_{j,q} + \omega_j, \ldots, \omega_M\right]^T, \end{aligned} \tag{16}$$

where $\boldsymbol{\beta}_{i,j,q} = \left[0, \ldots, 0, \beta_{i,q}, 0, \ldots, 0, \beta_{j,q}, 0, \ldots, 0\right]^T$. $\tilde{\beta}_{i,q}$ and $\tilde{\beta}_{j,q}$ are the complex Gaussian random variables with means $\beta_{i,q}$ and $\beta_{j,q}$, respectively.

The test statistics of the NCI-detector is the weighted quadratic sum of all observation data gathered by all receiving stations. For TDSRC, the test statistics can be regarded as the spatial energy accumulation of the target echo received by all radar stations. However, the observation vector of the DDSRC contains noise-only component, and thus results in the loss of the spatial power superposition of the jamming signal. Therefore, the jamming signal cannot be detected with an optimal performance. Fig. 5 shows a typical spatial power superposition for the TDSRC, DDSRC, and relative SRCs in the signal fusion center where the signal-to-noise ratio (SNR) is 10 dB and the jamming-to-noise ratio (JNR) is 13 dB. The SNR of kth target echo is defined as $\mathrm{SNR} = \boldsymbol{\alpha}_k^H \boldsymbol{\alpha}_k / (M \cdot \sigma)$ and the JNR of qth jamming signal is $\mathrm{JNR} = \boldsymbol{\beta}_q^H \boldsymbol{\beta}_q / (M \cdot \sigma)$. The power of a target can converge on the TDSRC, whereas that of a jamming cannot be gathered in an SRC. This phenomenon leads to the dispersion of spatial power.

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C. Case Study of the NCI Detector

The NCI detector is used for centralized target detection in multistatic radar with long baseline among radar stations. Given that the jamming signal retransmitted by the repeater

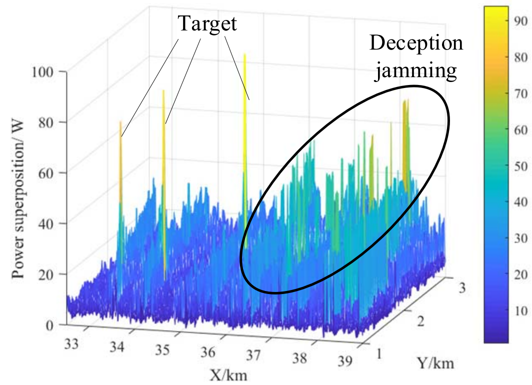

Fig. 5. Spatial power superposition in the scenario where SNR=10 dB and JNR=13 dB.

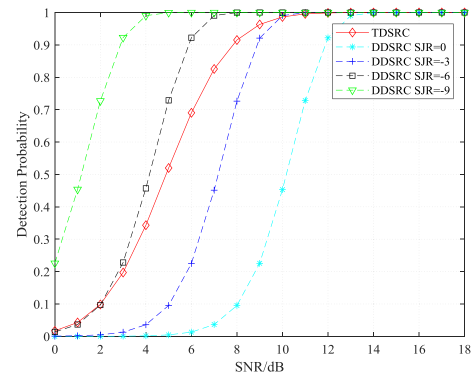

Fig. 6. Detection probabilities of TDSRC and DDSRC.

jammer is similar to the target echo in the time and frequency domains [27], the former can be mistakenly detected as the former by the NCI detector. However, the energy accumulation of the jamming signal and target echo is different due to the difference in their registration performances. In this subsection, we take the NCI detector as an example to provide a detailed analysis of the target echo and jamming signal.

Assuming that the complex amplitude of the target follows the complex Gaussian distribution with zero mean and variance λ^2, the NCI detector can be expressed as

$$\mathcal{F}_{NCI}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) = \frac{1}{\sigma^2}\mathbf{z}(\boldsymbol{\theta}_g)^H\mathbf{z}(\boldsymbol{\theta}_g) \underset{\mathrm{H}_0}{\overset{\mathrm{H}_1}{\gtrless}} \eta_{NCI}, \tag{17}$$

where η_{NCI} of the NCI detector. The test statistics follows the chi-square distribution with $2M$ degrees of freedom under hypothesis H_0. The false alarm probability is obtained as

$$P_{fa} = \exp\left(-\eta_{NCI}\right)\sum_{n=0}^{M-1}\frac{\left(\eta_{NCI}\right)^n}{\Gamma(n)}. \tag{18}$$

By substituting (13) to (17), the test statistics of a TDSRC can be written as

$$\mathcal{F}_{NCI}\left(\mathbf{z}_k^p(\boldsymbol{\theta}_k^T)\right) = \frac{1}{\sigma^2}\sum_{m=0}^{M-1}\left|\alpha_{m,k}+\omega_m\right|^2. \tag{19}$$

For the target model $\boldsymbol{\alpha}_k \sim \mathcal{CN}\left(\mathbf{0}, \lambda_k^2\mathbf{I}_M\right)$, the detection probability can be determined as

$$P_d = \exp\left(-\eta_{NCI}/\tilde{\sigma}_t^2\right)\sum_{n=0}^{M-1}\frac{\left(\eta_{NCI}/\tilde{\sigma}_t^2\right)^n}{\Gamma(n)}, \tag{20}$$

where $\tilde{\sigma}_t^2 = 1+\lambda_k^2/\sigma^2$.

The global test statistic of a DDSRC is expressed as

$$\mathcal{F}_{NCI}\left(\mathbf{z}(\boldsymbol{\theta}_{i,j,q}^{jam})\right) = \underbrace{\frac{1}{\sigma^2}\left|\beta_{i,q}+\omega_i\right|^2}_{\tilde{\beta}_{i,q}} + \underbrace{\frac{1}{\sigma^2}\left|\beta_{j,q}+\omega_j\right|^2}_{\tilde{\beta}_{j,q}} + \underbrace{\frac{1}{\sigma^2}\sum_{m=0,m\neq i,j}^{M}\left|\omega_m\right|^2}_{\tilde{\beta}_{\omega}}, \tag{21}$$

where $\tilde{\beta}_{i,q}$ and $\tilde{\beta}_{j,q}$ are complex Gaussian random variables with means $\beta_{i,q}$ and $\beta_{j,q}$, respectively. Given that the test statistics is the sum of the squares of Gaussian random variables with non-zero means, the test statistics follows the non-central chi-square distribution with 2M degrees of freedom and non-centrality parameter $\mu = \left|\beta_{i,q}\right|^2 + \left|\beta_{j,q}\right|^2$, which is denoted as $\chi_{2M}^{\dagger 2}(\mu)$ [37]. The probability distribution function (PDF) can be expressed in infinite series form as

$$f_{\chi_{2M}^{\dagger 2}(\mu)}(z) = \frac{z^{M-1}\exp\left[(z+\mu)/2\right]}{2^M}\sum_{k=0}^{\infty}\frac{(\mu z/4)^k}{k!\Gamma(M+k)}. \tag{22}$$

The probability of detecting the jamming signal detected as the target echo is obtained as

$$P_d^j = \int_{\eta_{NCI}}^{\infty} f_{\chi_{2M}^{\dagger 2}(\mu)}(z)dz. \tag{23}$$

The detection probability of TDSRC and DDSRC obtained by the NCI detector test algorithm is calculated using the Monte Carlo (MC) method. Fig. 6 shows that when the signal-to-jamming ratio (SJR) is 0 and -3 dB, the detection probability of the jamming signal is lower than that of the target echo, which implies that false targets may not be generated when the target is detected. When the SJR is smaller than -6 dB, the detection performance for the former increases. Many false targets will confuse the radar system under this situation.

The NCI-based detection algorithm can suppress deception jamming because the jamming signal is detected with a performance loss. However, the power of the jamming signal is usually larger than that of the target echo. As a result, the DDSRCs and its relative SRCs with no physical target can be mistakenly detected as the target; such a mistake leads many false targets during the NCI-based target detection.

IV. Joint Target Detection and Deception Jamming Suppression Method

As previously established, the DDSRCs and relative SRCs can be inaccurately detected during NCI detection. To solve this problem, a joint target detection and deception jamming suppression method is derived to eliminate the interference caused by the DDSRC and its relative SRCs.

The existence of the relative SRCs is disregarded. By taking the kth target and qth jamming signal as examples, the detection problem can be expressed as a multiple hypothesis test.

$$\mathbf{z}(\boldsymbol{\theta}_g) = \begin{cases} \boldsymbol{\omega} & H_0 \\ \boldsymbol{\alpha}_k + \boldsymbol{\omega} & H_1 \\ \boldsymbol{\beta}_{i,j,q} + \boldsymbol{\omega} & H_2 \end{cases} \tag{24}$$

where H_0 is the hypothesis that the SRC contains only noise components, H_1 is the hypothesis that the SRC belongs to TDSRC, which contains target of interest, and H_2 is the hypothesis that the SRC belongs to DDSRC. The PDFs of these hypotheses are given accordingly.

A. Discrimination of the DDSRC

To eliminate the interference caused by DDSRCs, the DDSRCs must be discriminated from TDSRCs and noise-only SRCs. The observation vector follows the zero-mean Gauss distribution under H_1 and H_0. The PDF is given as

$$p\left(\mathbf{z}(\boldsymbol{\theta}_g)|H_b\right) = \prod_{m=0}^{M} \frac{1}{\sqrt{2\pi}\tilde{\sigma}} \exp\left\{-\frac{\left|z_m(\boldsymbol{\theta}_g)\right|^2}{2\tilde{\sigma}^2}\right\} \quad b = 0, 1, \tag{25}$$

where $\tilde{\sigma}^2$ represents the estimation of the noise level under H_0 and target power under H_2. In practice, $\tilde{\sigma}^2$ need to be trained using the observation vector of the probed SRC. Under H_2, only two elements are present in the observation vector collected from the deception jamming. The corresponding PDF is expressed as

$$\begin{aligned} p&\left(\mathbf{z}\left(\boldsymbol{\theta}_g\right)|\boldsymbol{\beta}_{i,j,q}, H_2\right) \\ &= \prod_{m=i,j} \frac{1}{\sqrt{2\pi}\tilde{\sigma}} \exp\left\{-\frac{\left|z_m(\boldsymbol{\theta}_g) - \beta_{m,q}\right|^2}{2\tilde{\sigma}^2}\right\} \\ &\quad \times \prod_{m=0, m\neq i,j}^{M} \frac{1}{\sqrt{2\pi}\tilde{\sigma}} \exp\left\{-\frac{\left|z_m(\boldsymbol{\theta}_g)\right|^2}{2\tilde{\sigma}^2}\right\}. \end{aligned} \tag{26}$$

The discrimination problem can be written as a binary hypothesis test. In accordance with the adaptive match filter algorithm, the likelihood ratio test is defined as

$$\begin{aligned} L_D&\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) \\ &= \frac{\max_{\boldsymbol{\beta}_{i,j,q}} p\left(\mathbf{z}(\boldsymbol{\theta}_g)|\boldsymbol{\beta}_{i,j,q}, \sigma, H_2\right)}{p\left(\mathbf{z}(\boldsymbol{\theta}_g)|\tilde{\sigma}, H_b\right)} \\ &= \prod_{m=i,j} \exp\left\{-\frac{\left|z_m(\boldsymbol{\theta}_g) - \beta_{m,q}\right|^2}{2\tilde{\sigma}^2} + \frac{\left|z_m(\boldsymbol{\theta}_g)\right|^2}{2\tilde{\sigma}^2}\right\} \underset{H_0+H_1}{\overset{H_2}{\gtrless}} \gamma. \end{aligned} \tag{27}$$

In (27), $\boldsymbol{\beta}_{i,j,q}$ is an unknown parameter that must be estimated through maximum likelihood estimation (MLE).

$$\hat{\boldsymbol{\beta}}_{i,j,q} = \arg\max_{\boldsymbol{\beta}_{i,j,q}} \ln p\left(\mathbf{z}\left(\boldsymbol{\theta}_g\right)|\boldsymbol{\beta}_{i,j,q}, \sigma, H_2\right) \tag{28}$$

By simplifying (28), the likelihood function can be expressed as

$$\hat{\boldsymbol{\beta}}_{i,j,q} = \arg\max_{\boldsymbol{\beta}_{i,j,q}} \left\{\sum_{m=i,j} -\left|z_m(\boldsymbol{\theta}_g) - \beta_{m,q}\right|^2 + \left|z_m(\boldsymbol{\theta}_g)\right|^2\right\}. \tag{29}$$

The likelihood function is maximized by assuming that $\boldsymbol{\beta}_{i,j,q} = \hat{\boldsymbol{\beta}}_{i,j,q}$, where $\hat{\boldsymbol{\beta}}_{i,j,q}$ is given as

$$\hat{\boldsymbol{\beta}}_{i,j,q} = \left[0, \ldots, z_i(\boldsymbol{\theta}_g), \ldots, z_j(\boldsymbol{\theta}_g), \ldots, 0\right]^T. \tag{30}$$

By substituting (30) to (27), the likelihood ratio function can be expressed as

$$L_D\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) \propto \frac{\left|z_i(\boldsymbol{\theta}_g)\right|^2 + \left|z_j(\boldsymbol{\theta}_g)\right|^2}{\tilde{\sigma}^2} \underset{H_0+H_1}{\overset{H_2}{\gtrless}} \gamma', \tag{31}$$

where i and j are unknown with C_M^2 possibilities. As a result, the combination of i and j that will maximize the objective function should be selected.

$$[\hat{i}, \hat{j}] = \arg\max_{i,j} \left\{\sum_{m=i,j} \left|z_m(\boldsymbol{\theta}_g)\right|^2\right\} \tag{32}$$

The estimated jamming signal amplitude is equivalent to the two largest elements in the observation vector. Inserting (32) into (31), we have

$$D\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) = \frac{\max_{i,j}\left\{\sum_{m=i,j} \left|z_m(\boldsymbol{\theta}_g)\right|^2\right\}}{\tilde{\sigma}^2} \underset{H_0+H_1}{\overset{H_2}{\gtrless}} \gamma'. \tag{33}$$

Given that $z_{\hat{i},g}$ and $z_{\hat{j},g}$ are the two largest elements in the observation vector, we can obtain the joint PDF of them with several derivations (also see in Appendix).

$$\begin{aligned} &f_{\tilde{z}_{\hat{i},g}, \tilde{z}_{\hat{j},g}}(z_1, z_2|H_0 + H_1) \\ &= M(M-1)\exp(-z_2) f(z_1, z_2)\left[\int_0^z f(t, z_2)dt\right]^{M-2} \end{aligned} \tag{34}$$

where

$$f(z_1, z_2) = \begin{cases} \exp(-z_1) & 0 < z \le z_2 \\ 0 & \textit{otherwise}. \end{cases} \tag{35}$$

The probability of detecting DDSRC as noise-only or target-presence SRC is determined as

$$\begin{aligned} P_e &= P\{H_2|H_0 + H_1\} \\ &= \iint_{z_1+z_2>\gamma'} f_{z_{\hat{j},g}, z_{\hat{i},g}}(z_1, z_2|H_0 + H_1)dz_1dz_2 \\ &= 1 - \iint_{z_1+z_2\le\gamma'} f_{z_{\hat{j},g}, z_{\hat{i},g}}(z_1, z_2|H_0 + H_1)dz_1dz_2. \end{aligned} \tag{36}$$

B. Interference Cancellation Algorithm

In the previous subsection, we focus on discriminating DDSRC from TDSRC and noise-only SRC but disregard the existence of the relative SRCs. The relative SRCs of TDSRC and DDSRC can result in numerous false alarms, and thus must be taken into consideration.

To eliminate the interference of the relative SRC as well as DDSRC, the detector of NCI-based algorithm is modified.

$$\max_{c(\boldsymbol{\theta}_g)\in\mathbf{O}} \mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) \underset{H_0+H_2}{\overset{H_1}{\gtrless}} \eta_{IC} \tag{37}$$

where

$$\mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) = \mathcal{F}_{NCI}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) - M_{DC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) - M_{RC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right), \tag{38}$$

where $\mathbf{O} \subseteq \boldsymbol{\Omega}$ represents the set of the untested SRC. $\mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ is the test statistic of interference cancellation (IC) algorithm. $M_{DC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ is designed to eliminate the interference of DDSRC. In accordance with (33), the modified term can be defined as

$$M_{DC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) = \begin{cases} D\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) & D\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) > \gamma' \\ 0 & \textit{otherwise}. \end{cases} \tag{39}$$

The term $M_{RC}\left(\mathbf{z}\left(\boldsymbol{\theta}_g\right)\right)$ is derived to cancel the interference of the relative SRC. Considering that the relative SRC of TDSRC and DDSRC can be inaccurately detected, the modified term is given as

$$M_{RC}(\mathbf{z}(\boldsymbol{\theta}_g)) = \sum_{m=0}^{M-1}\sum_{l^T=0}^{L^T-1} M^T_{m,l^T}(\mathbf{z}(\boldsymbol{\theta}_g)) + \sum_{m=0}^{M-1}\sum_{l^J=0}^{L^J-1} M^J_{m,l^J}(\mathbf{z}(\boldsymbol{\theta}_g)), \tag{40}$$

where L^T and L^J are the numbers of declared TDSRC and DDSRC. $M^T_{m,l^T}\left(\mathbf{z}\left(\boldsymbol{\theta}_g\right)\right)$ and $M^J_{m,l^J}\left(\mathbf{z}\left(\boldsymbol{\theta}_g\right)\right)$ are used for the relative SRCs of the targets and jamming signals, respectively. The modified term related to the l^Tth declared TDSRC in the mth receiving station can be expressed as

$$M^T_{m,l^T}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) = \begin{cases} \left|z_m(\boldsymbol{\theta}_g)\right|^2/\sigma^2 & c(\boldsymbol{\theta}_g) \in \mathbb{B}_m(\boldsymbol{\theta}^T_{l^T}) \\ 0 & \textit{otherwise}, \end{cases} \tag{41}$$

where $\mathbb{B}_m(\boldsymbol{\theta}^T_{l^T})$ is the set of the SRCs affected by TDSRC $c(\boldsymbol{\theta}^T_{l^T})$ in the mth receiving station. This term can be expressed as $\mathbb{B}_m(\boldsymbol{\theta}^T_{l^T}) = \boldsymbol{\Psi}_m(\boldsymbol{\theta}^T_{l^T})$.

Unlike the relative SRCs of TDSRCs, those of DDSRCs from only two branches can be influenced by the jamming signal. The corresponding modified term is defined as

$$M^J_{m,l^J}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) = \begin{cases} \left|z_m(\boldsymbol{\theta}_g)\right|^2/\sigma^2 & c(\boldsymbol{\theta}_g) \in \mathbb{A}_m(\boldsymbol{\theta}^J_{l^J}) \\ 0 & \textit{otherwise}, \end{cases} \tag{42}$$

where $\mathbb{A}_m(\boldsymbol{\theta}^J_{l^J})$ is the set of the SRCs affected by DDSRC $c\left(\boldsymbol{\theta}^J_{l^J}\right)$ in the mth receiving station. Considering that not all relative SRCs can be influenced by the DDSRC, this term is expressed as

$$\mathbb{A}_m(\boldsymbol{\theta}^J_{l^J}) = \begin{cases} \boldsymbol{\Psi}_m(\boldsymbol{\theta}^J_{l^J}) & m = \hat{i}\, or\, \hat{j} \\ \emptyset & \textit{otherwise}. \end{cases} \tag{43}$$

Algorithm 1: The Summary of IC Algorithm

Form the original set of TDSRCs $\mathbb{T} = \emptyset$, the set of DDSRCs $\mathbb{D} = \emptyset$, and the set of probed SRCs $\mathbb{O} = \boldsymbol{\Omega}$.
for $g = 1, 2, \ldots, G$ **do**
  Obtain the SRC maximizing the statistical $c(\hat{\boldsymbol{\theta}}_g) = \arg\max_{c(\boldsymbol{\theta}_g)\in\mathbb{O}} \mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$.
  $\mathbb{O} = \mathbb{O} - c(\hat{\boldsymbol{\theta}})$.
  if $D\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) > \gamma'$
    if $\max_{c(\boldsymbol{\theta}_g)\in\mathbb{O}} \mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) > \eta_{IC}$
      Add the SRC $c(\hat{\boldsymbol{\theta}})$ to the set $\mathbb{T}$.
    else
      Add the SRC $c(\hat{\boldsymbol{\theta}})$ to the set $\mathbb{D}$.
    end
  else
    if $\max_{c(\boldsymbol{\theta}_g)\in\mathbb{O}} \mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) > \eta_{IC}$
      Add the SRC $c(\hat{\boldsymbol{\theta}})$ to the set $\mathbb{T}$.
    end
  end
end
Output the set of TDSRCs $\mathbb{T}$.

The threshold η_{IC} is the function of the false alarm probability, number of declared target, and type of SRC (i.e., TDSRC or DDSRC). Given that the test statistics might change after an SRC is declared, the stationary threshold cannot be matched with the modified statistics. Therefore, the detection threshold should be adjusted accordingly. Given that the statistics of the IC algorithm equals to the weighted quadratic sum of all observation data that are not affected by declared DDSRC and detected TDSRC, $\mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ can be regarded as a random variable that follows chi-square distribution with m_{IC} degrees of freedom under H_0. The threshold is then obtained as

$$P_{fa} = \exp\left(-\eta_{IC}\right) \sum_{n=0}^{m_K/2-1} \frac{(\eta_R)^n}{\Gamma(n)}, \tag{44}$$

where

$$m_{IC} = \begin{cases} 2M - 4 - 2F_{M,L^T,L^J}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) & D\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) > \gamma' \\ 2M - 2F_{M,L^T,L^J}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) & \textit{otherwise}, \end{cases} \tag{45}$$

$$F_{M,L^T,L^J}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right) = \sum_{m=0}^{M-1}\sum_{l^T=0}^{L^T-1} u^T_{m,l^T} + \sum_{m=0}^{M-1}\sum_{l^J=0}^{L^J-1} u^J_{m,l^J}. \tag{46}$$

In (46), $u^T_{m,l^T} = 1$ if $M^T_{m,l^T}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ is non-zero. otherwise, $u^T_{m,l^T} = 0$. Similarly, $u^J_{m,l^J} = 1$ if $M^J_{m,l^J}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ is non-zero. Otherwise, $u^J_{m,l^J} = 0$.

In summary, all SRCs in the common surveillance region must be tested. When an SRC is tested, target detection and deception jamming suppression is simultaneously implemented. To eliminate the interference of DDSRC and relative SRCs, several modified terms are subtracted from the NCI-based test statistics. Modified terms containing the information of the declared SRC are designed to make the proposed

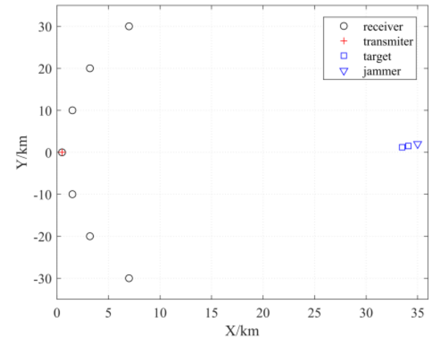

Fig. 7. The placement of the MSRS.

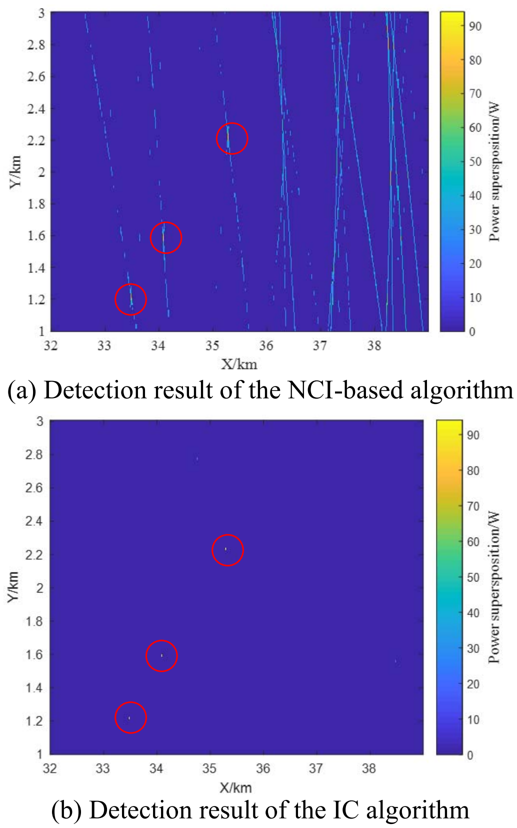

Fig. 8. Detection results in the scenario with SNR=10 dB and JNR=13 dB.

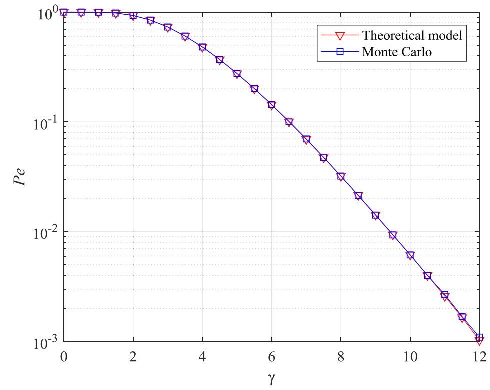

Fig. 9. P_e versus the discriminator test threshold.

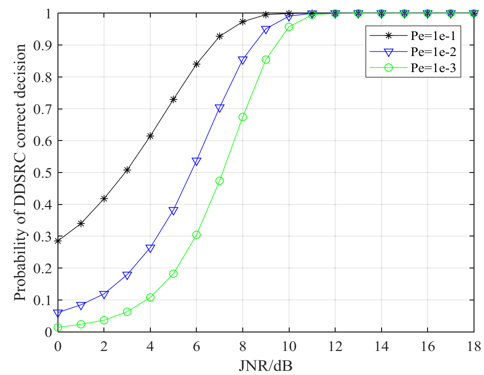

Fig. 10. Discrimination performance of the DDSRC discriminator.

detector reject DDSRC and the relative SRC. Given that the test statistics might change as soon as an SRC is tested, the threshold should be adjusted accordingly. The pseudo code of this algorithm is given in Algorithm 1.

Computational Complexity Analysis: From the summary of IC algorithm, the computational costs of the algorithm is dominated by the calculation of the test statistics $\mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ and the numbers of SRCs in the common surveillance region. Assume that the received signals has been transformed into the observation vector by (7)-(9), the computational complexity of computing $M_{DC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ and $\mathcal{F}_{NCI}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ is $O\left(M\right)$. The modified term $M_{RC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ leads to a complexity with order $O\left(N_T M\right) + O\left(N_J M\right)$ where N_T and N_J are the number of TDSRCs and DDSRCs respectively. Besides, we must maximize $\mathcal{F}_{IC}\left(\mathbf{z}(\boldsymbol{\theta}_g)\right)$ G times to test all the SRCs, then the total computation complexity of IC algorithm is $O\left(N_T G^2 M\right) + O\left(N_J G^2 M\right)$.

V. Simulation Results

In this section, we first simulate both the IC algorithm and the NCI-based detection algorithm in the presence of deception and compare the IC algorithm with the NCI-based algorithm. Because the performance of the proposed IC algorithm is related to the DDSRC discrimination, the performance of DDSRC discrimination need to be analyzed firstly. And then the performance of IC algorithm is given in terms of target detection and deception suppression respectively.

A. Simulation of IC Algorithm

The scenario where three targets, a jammer, and an MSRS exist is considered to simulate the IC algorithm. The deployment of MSRS, which involves a transmitter and seven receivers, is illustrated in Fig. 7. The transmitter is placed at (0, 0) km, whereas and the receivers are placed at (4.4, -30), (1.8, -20), (0.8, -10), (0, 0), (0.8, 10), (1.8, 20), and (4.4, 30) km. In the common surveillance area, the three targets are respectively located at (35.3, 2.2), (34.1, 1.6), and (33.5, 1.2) km. The self-defensive repeater jammer is designed to be carried by the first target. Therefore, the assumption is that all receivers can receive the signal retransmitted by the jammer.

The detection results are depicted in Fig. 8, where the bright spot represents the target detected by radar system. The bright spots trapped in cycle are the physical targets, whereas

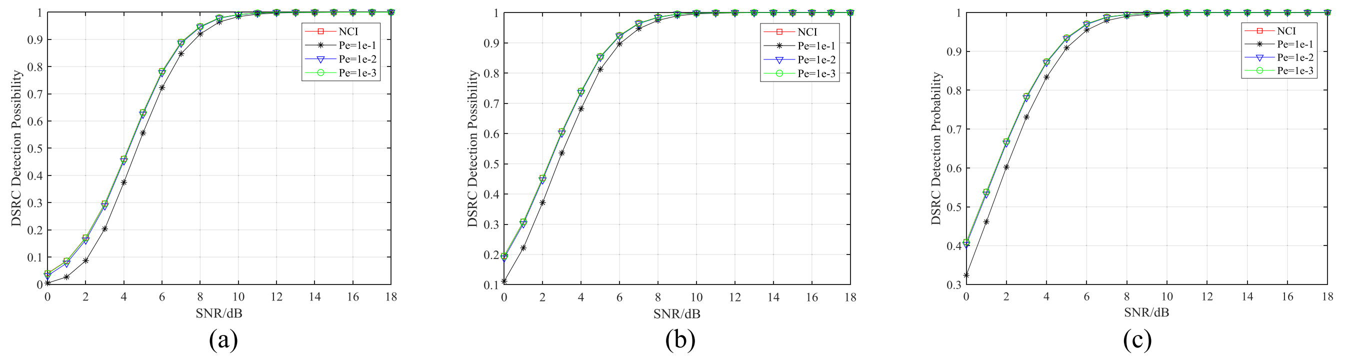

Fig. 11. Detection probabilities of TDSRC (a) $P_{fa} = 10^{-5}$, (b) $P_{fa} = 10^{-3}$, (c) $P_{fa} = 10^{-2}$.

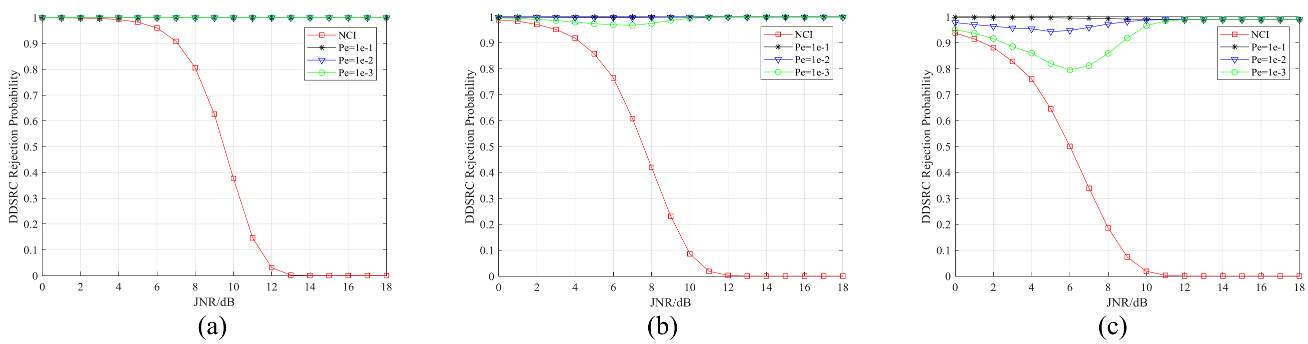

Fig. 12. Rejection probabilities of DDSRC (a) $P_{fa} = 10^{-5}$, (b) $P_{fa} = 10^{-3}$, (c) $P_{fa} = 10^{-2}$.

the other spots represent the false ones. Fig. 8(a) shows that the NCI-based detection algorithm generates many false targets in the presence of deception jamming. In comparison, the IC algorithm detects all physical targets correctly without generating any false target. In conclusion, the IC algorithm can execute effective detection in the environment with deception jamming.

B. Performance Analysis

As previously mentioned, the discrimination of the DDSRC is the prerequisite of the proposed algorithm. The discrimination performance of the DDSRC is crucial to the final decision of the IC algorithm. Therefore, we analyze the performance of the DDSRC discriminator before performing the final detection.

The threshold of the discriminator is designed on the basis of the principle of constant misjudgment probability. Fig. 9 shows that the thresholds obtained using (36) coincide with that obtained through the MC approach.

The probability of DDSRC to obtain a correct decision $P\{H_2|H_2\}$ is used to evaluated the performance of the discriminator. Fig. 10 depicts the performance of depicts the probability of the discriminator under different misjudgment probabilities (P_e). The performance of the discriminator deteriorates as the JNR decreases. In addition, a smaller misjudgment probability signifies a larger discrimination threshold. Therefore, the probability of DDSRC to achieve a correct decision is high if the misjudgment probability is large.

The result indicates that proposed IC algorithm can suppress deception jamming while detecting the target. The performance of the IC algorithm is then evaluated using two measurements.

The probability of TDSRC detection P_d:P_d is defined as the probability that the target contained by the TDSRC is detected. This probability, which can be written as $P_d = P\{H_1|H_1\}$, is the first measurement used to assess the detection performance of the IC algorithm.

The second measurement is the probability of DDSRC rejection P_r:P_r, which is defined as the probability that the DDSRC containing no targets is rejected. This indicator, which can be expressed as $P_r = 1 - P\{H_1|H_2\}$, can describe the ability to suppress deception jamming.

Fig. 11 illustrates the probability of target detection of the IC and NCI-based detection algorithms under different P_e values. A smaller P_e means that less TDSRCs are treated as DDSRCs. The detection performance slightly declines as P_e increases. This phenomenon occurs because the IC algorithm will remove the two largest elements from the observation vector, which then leads to energy loss if the TDSRC is mistaken for DDSRC. Given that the performance degradation is minimal, the characteristic performance curve at $P_e = 10^{-3}$ is coincident.

Fig. 12(a) shows that the rejection probability of the NCI detection algorithm decreases as JNR increases. By contrast, the IC algorithm maintains a 100% probability because of its ability to discriminate and eliminate deception jamming when the JNR is sufficiently high and treat the jamming as noise when the JNR becomes sufficiently low to be rejected. The result of comparing Figs. 12(a) and 12(b) suggests that the curve of the rejection probability slightly declines at $P_e = 10^{-3}$. When JNR = 6 dB, the rejection probability drops. In Fig 12(c), This degradation becomes severe in Fig. 12(c); the rejection probability decreases to 80% at $P_e = 10^{-3}$ when JNR = 6 dB. The jamming suppression performance is influenced by P_e and P_{fa}. As P_{fa} increases, the jamming signal with low JNR can be detected. If the JNR is not sufficiently large to distinguish the jamming signal from other signals, the signal might be rejected with a small probability. To obtain a satisfactory deception jamming suppression performance, the P_e that matches with P_{fa} should be selected to maintain the rejection probability at a high level.

VI. Conclusion

In this study, we investigate the target detection performance of multistatic radar in the presence of deception jamming. We analyze the characteristic of TDSRC and DDSRC. The results reveal that DDSRC is still detected by mistake, and thus leads to several false targets. To address this problem, which cannot be resolved by the NCI-based detection algorithm, we propose the joint target detection and deception jamming suppression method. We derive a discriminator to distinguish DDSRC from the others and then eliminate the interference of DDSRC and relative SRCs during target detection to simultaneously detect targets and suppress deception jamming. The discrimination between actual and false targets is no longer required after the implementation of the IC algorithm. The simulation results show that the proposed algorithm exhibits better jamming suppression performance than the NCI detector with a slight loss of target detection performance.

Appendix
Derivation of (34)

Under H_0 or H_1, $\tilde{z}_{m,g}$ follows the exponential distribution with a PDF expressed as

$$f_{\tilde{z}_{m,g}}(z|H_0+H_1) = \exp(-z). \tag{47}$$

Considering that $\tilde{z}_{\hat{i},g}$ is the maximum of M exponential random variables, the corresponding PDF can be expressed as

$$f_{\tilde{z}_{\hat{i},g}}(z|H_0+H_1) = M\exp(-z)\left[1-\exp(-z)\right]^{M-1}, \tag{48}$$

Assume that $\tilde{z}_{j,g}$ is a smaller exponential random variables than $\tilde{z}_{\hat{i},g}$, the PDF of $\tilde{z}_{i,g}$ can be obtained as

$$f_{\tilde{z}_{j,g}|\tilde{z}_{\hat{i},g}}(z|z_2, H_0+H_1) = \frac{f(z,z_2)}{1-\exp(-z_2)}, \tag{49}$$

where

$$f(z_1,z_2) = \begin{cases} \exp(-z_1) & 0 < z \le z_2 \\ 0 & \textit{otherwise}. \end{cases} \tag{50}$$

Given $\tilde{z}_{\hat{j},g}$ is larger than all observation data except $\tilde{z}_{\hat{i},g}$, the cumulative distribution function of $\tilde{z}_{\hat{j},g}$ given $\tilde{z}_{\hat{i},g}$ is

$$F_{\tilde{z}_{\hat{j},g}|\tilde{z}_{\hat{i},g}}(z_1|z_2) = \frac{\left[\int_0^{z_1} f(t,z_2)\mathrm{d}t\right]^{M-1}}{\left[1-\exp(-z_2)\right]^{M-1}}. \tag{51}$$

Then, the PDF of $\tilde{z}_{\hat{j},g}$ given $\tilde{z}_{\hat{i},g}$ can be given as

$$\begin{aligned} f_{\tilde{z}_{\hat{i},g}|\tilde{z}_{\hat{j},g}}&(z_1|z_2, H_0+H_1) \\ &= \mathrm{d}F_{\tilde{z}_{\hat{j},g}|\tilde{z}_{\hat{i},g}}(z_1|z_2)/\mathrm{d}z_1 \\ &= \frac{(M-1)\,f(z_1,z_2)\left[\int_0^{z_1} f(t,z_2)dt\right]^{M-2}}{\left[1-\exp(-z_2)\right]^{M-1}}. \end{aligned} \tag{52}$$

Hence, the joint PDF probability of and $z_{\hat{j},g}$ can expressed as

$$\begin{aligned} f_{\tilde{z}_{\hat{i},g},\tilde{z}_{\hat{j},g}}&(z_1,z_2|H_0+H_1) \\ &= f_{\tilde{z}_{\hat{i},g},\tilde{z}_{\hat{j},g}}(z_1|z_2,H_0+H_1)\cdot f_{\tilde{z}_{\hat{i},g}}(z_2|H_0+H_1) \\ &= \frac{(M-1)\,f(z_1,z_2)\left[\int_0^{z} f(t,z_2)dt\right]^{M-2}}{\left[1-\exp(-z_2)\right]^{M-1}} \\ &\quad\times M\exp(-z_2)\left[1-\exp(-z_2)\right]^{M-1} \\ &= M(M-1)\exp(-z_2)\,f(z_1,z_2)\left[\int_0^{z} f(t,z_2)dt\right]^{M-2}. \end{aligned} \tag{53}$$

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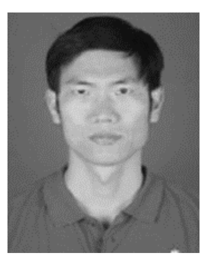

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